Florida Mobile Lidar Projects, Nov 2023 – Nov 2026

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Summary

No Florida mobile lidar project in this window has public documentation showing it meets the ASPRS Positional Accuracy Standards, Edition 2, Version 2 (2024). There is no document called "ASPRS State Lidar Standards"; Edition 2 v2 is the standard used here.

- **Why:** FDOT work is specified to the FDOT Surveying and Mapping Handbook (02-28-2025). It tests mobile lidar under NSSDA at 95% confidence with at least 20 checkpoints. ASPRS requires at least 30 checkpoints and reports RMSE.
- **What was found:** about 15 mobile lidar efforts tied to Florida in or near the window. Many dates come only from vendor marketing pages and are unverified.
- **Datasets:** almost none of the point clouds are public. Florida has no statewide registry of mobile lidar projects. Public lidar portals hold airborne data almost exclusively.
- **Only public accuracy report:** Wood's 2019 memo on Monroe County roads, outside the window. It used 108 checkpoints and found a road RMSE of 0.09 ft, judged acceptable for flood-vulnerability work.

Verdict codes used in the tables:

Code	Meaning
A	Documented as meeting ASPRS
F	Meets FDOT mobile lidar standard; ASPRS not documented
U	Fitness-for-use assessment only
X	Undetermined; no public accuracy documentation

State projects

Nine state-level entries were found; none has a public dataset or a published accuracy report. Request data from the owner under Chapter 119, F.S.

Project	Owner	Firm	Dates	System, extent, class	Request data from	Verdict
Outdoor Advertising Program statewide inventory	FDOT Central Office	WGI	Five-year contract; award announced 2026. Start date and contract number not found	Mobile lidar and 360° imagery; 13,000+ mi; 9,500+ billboards; 100+ TB. Class not published	FDOT Outdoor Advertising / Right of Way	X
SR 570 Polk Parkway resurfacing, MP 8–17.2, Polk Co.	Florida's Turnpike Enterprise	DRMP	Undated; likely 2024 or later (inference)	RIEGL VMX-2HA with 360° camera; 6.6 mi stated (mileposts imply 9.2 mi); Type A	Turnpike Surveying & Mapping, Ocoee	F (vendor claim only)
I-175/SR 594, St. Petersburg, concrete rehabilitation	FDOT District 7	DRMP	Undated	RIEGL VMX-2HA; mileage and class not stated	District 7 survey office, Tampa	X
SR 30A (US 98) Panama City Beach Parkway, Bay Co.	FDOT District 3	DRMP	Undated	RIEGL VMX-2HA; 0.95 mi; 62+ targets; class not stated	District 3 survey office, Chipley	X
Tampa Westshore Interchange, design-build phase	FDOT District 7	DRMP	Called "most recent" on Oct 16, 2024	Mobile lidar; details not published	District 7 survey office or design-build contractor	X
Turnpike Mainline resurfacing, St. Lucie and Indian River Cos.	Florida's Turnpike Enterprise	DRMP	Undated; possibly before the window	Mobile lidar; about 9 mi	Turnpike Surveying & Mapping	X

Project	Owner	Firm	Dates	System, extent, class	Request data from	Verdict
Lovers Key State Park restoration, Lee Co.	FDEP	DRMP	Undated	RIEGL VMX-2HA; combined with static, drone and bathymetric survey	FDEP Division of Recreation & Parks	X
Turnpike surveying and mapping continuing services contract	Florida's Turnpike Enterprise	Various	Active advertisement	Contract vehicle; scope includes mobile lidar (Activity 30) per task order	Turnpike Surveying & Mapping	Not applicable
Roadway Characteristics Inventory pilot (context)	FDOT Transportation Data & Analytics	GPI Geospatial	Reported April 2021; outside the window	RIEGL VMX-450; about 28 centerline mi; ±0.25 ft thresholds. No statewide follow-on found	FDOT Central Office	U

County and municipal projects

The Monroe County municipal survey and the City of Sarasota mapping are the two largest local projects in the window. Only Monroe County offers any public view of the data.

Project	Owner	Firm	Dates	System, extent, class	Dataset access	Verdict
Mobile lidar for five Keys municipalities (Key West, Marathon, Islamorada, Layton, Key Colony Beach)	Monroe County	WSP USA (formerly Wood)	Key West collected Sept 18 – Oct 6, 2023. Contract extended to Jan 30, 2025	Up to 249 mi scoped; Key West 72 mi (2023 notice) or 93 mi (2020 budget). Sensor and class not published	Road elevation viewer at keysroadsplan.com. Point cloud by request to Monroe County Office of Sustainability	X
Keys Roads Plan municipal vulnerability analysis, including the Islamorada Roads Elevation Study	Monroe County and municipalities	HDR (analysis), WSP (lidar)	HDR contract extended to Apr 30, 2026	Uses the municipal lidar above	Interactive maps at keysroadsplan.com	X
Monroe County roads mobile lidar (context)	Monroe County	Michael Baker (collection), Wood (checks)	Accuracy fieldwork June 25 – July 2, 2019; outside the window	311 mi; 108 checkpoints; road RMSE 0.09 ft; floor-elevation RMSE 0.15 ft	Accuracy memo public; data by request	U
Citywide topographic and Sarasota Bay resiliency mapping	City of Sarasota (FDEP Resilient Florida grant)	WGI	One-year contract approved Jan 21; year not shown, most likely 2025. Results presented in 2026	Aerial lidar plus mobile lidar of about 285 centerline mi (city page says about 330 mi); 700+ control points	Not public. Contact City Engineer or City Surveyor	X
Sand Key Park improvements	Pinellas County	DRMP	Undated	Mobile scan of park roads and parking	Not public. Pinellas County	X

Excluded: North Lauderdale's 2025 citywide sidewalk survey (109 mi) used handheld phone lidar, which is not survey-grade mobile mapping.

Private, institutional and rail projects

Only three entries were found, and only one clearly falls in the window. Private owners have no public records obligation.

Project	Owner	Firm	Dates	System, extent	Access	Verdict
Brightline West Palm Beach to Cocoa hi-rail corridor mapping	Brightline / HSR Constructors	WGI	Contract May 2019; collection almost certainly before the window	Hi-rail mobile lidar; about 130 mi	Private	X
Brightline locomotive lidar lab	Brightline	Sotereon.AI	Announced Nov 12, 2025	Sensors on two locomotives for corridor monitoring; not a survey product	Private	Not applicable
University of Florida campus lidar	University of Florida	DRMP	Undated	Mobile and drone lidar; 68 mi stated (page also says 20 mi)	Request from UF; Chapter 119 applies	X

Not found in the window: vehicle-based mobile lidar after Hurricanes Ian, Idalia, Helene or Milton; expressway authority, port or airport projects; Cyclomedia contracts with Florida counties. Post-storm lidar located was airborne.

SOPs and standards

The FDOT Surveying and Mapping Handbook is the governing specification for nearly all of this work. Project-specific procedures and survey reports are not published.

Document	Where	Key requirements
FDOT Surveying and Mapping Handbook, 02-28-2025	FDOT Documents & Publications	Accuracy relative to control at 95% confidence: Type A 0.06 ft, Type B 0.10 ft (horizontal and vertical), Type C per scope. Checkpoints every 1,000 ft or less (A). Point density 20 pts/ft² (A), 10 (B). NSSDA test with at least 20 checkpoints. Signed and sealed survey report stating the Type achieved
Terrestrial Mobile Lidar General Scope Guidelines (2014) and Staff Hour Guidelines (2015)	FDOT Documents & Publications	Scope and staff-hour templates for mobile lidar tasks (Activity 30)
FDOT Surveying and Mapping Procedure 550-030-101	FDOT Documents & Publications	Governs department surveying and mapping; the Handbook implements it
Rule 5J-17, F.A.C. (Standards of Practice)	flrules.org	Survey reports carry accuracy statements and are signed and sealed
ASPRS Positional Accuracy Standards, Edition 2, Version 2 (2024)	ASPRS announcement	RMSE only; at least 30 checkpoints, up to 120; checkpoint survey error counted in product accuracy; Mapping with Lidar addendum
ASPRS LAS 1.4	asprs.org	Point cloud file format; FDOT requires the latest LAS version
NCHRP Report 748 (2013)	onlinepubs.trb.org	Mobile lidar guidance referenced by FDOT

The Handbook values were read from PDF text extraction. Check them against the PDF table before using them in a contract.

ASPRS compliance determination

No project in the window is documented as meeting ASPRS. A compliant FDOT survey is not automatically ASPRS-compliant, because the two frameworks test and report accuracy differently.

	FDOT Handbook	ASPRS Edition 2 v2
Statistic	95% confidence level (NSSDA)	RMSE only
Minimum checkpoints	20	30
Checkpoint survey error	Not added	Added to product accuracy

For normally distributed, bias-free errors, vertical RMSE is about the 95% value divided by 1.96. Type A's 0.06 ft implies an RMSE near 0.031 ft; Type B's 0.10 ft implies about 0.051 ft. A Type A survey would likely perform at an ASPRS-equivalent level, but that is not documented unless it used 30 checkpoints and reported RMSE.

Project	Verdict	Evidence
FDOT Outdoor Advertising (WGI)	X	No class or checkpoint data published
SR 570 Polk Parkway (DRMP)	F	Vendor states Type A per FDOT guidelines; no checkpoint report public
Other DRMP state projects	X	No accuracy data; Panama City Beach lists 62+ targets, which is a control count
Keys municipalities (WSP) and Keys Roads Plan	X	No accuracy memo found for the 2023–25 dataset
Monroe County roads, 2019 (context)	U	108 checkpoints, RMSE 0.09 ft; no ASPRS class claimed
City of Sarasota (WGI)	X	700+ control points reported; no checkpoint RMSE public
Sand Key Park, Brightline, UF campus	X	Marketing pages only

FDOT deliverables are in ASPRS LAS format, so format compliance is likely but was not verified.

How to request datasets and reports

Request non-public data under Florida's public records law, Chapter 119, F.S. Ask for the point clouds and the signed and sealed survey report with its checkpoint table.

Owner	Where to send the request	What to cite or ask for
FDOT districts and Turnpike	District Surveying and Mapping Office; contacts on the FDOT Documents & Publications page	Financial Project ID, project limits, LAS point clouds, sealed survey report, QA/QC plan, boresight records
FDOT Central Office	FDOT public records request portal	Outdoor Advertising contract and scope with WGI
Monroe County	Office of Sustainability; Keys Roads Plan team, 305-453-8774	WSP contract and amendments 1–4, deliverables, any accuracy memo for the municipal lidar
Keys municipalities	City or village clerk	Local copies of the lidar deliverables
City of Sarasota	City Clerk public records portal; Public Works	WGI survey report and point cloud
FDEP	Office of Resilience and Coastal Protection	Resilient Florida grant deliverables (Sarasota, Keys)
University of Florida	UF Public Records	Campus lidar deliverables
Private owners (Brightline, WGI)	Voluntary request only	Copies may sit in FDOT or county permitting files

Public lidar repositories hold airborne data: NOAA Digital Coast, USGS 3DEP, OpenTopography and the Florida GIO lidar pages.

Gaps and recommendations

This inventory is incomplete: absence here means not found, not that no project exists.

Gaps and unverified items

- All DRMP entries and WGI's Outdoor Advertising and Brightline entries rest on vendor pages with no dates, contract numbers or accuracy reports.

- Agenda searches across all 67 counties were not completed, nor were searches of expressway authorities, ports and airports.
- Key West mileage conflicts: 72 mi or 93 mi.
- Sarasota extent conflicts: 285 or 330 mi; 8,400, 15,400 or 84,100 acres.
- SR 570 length conflicts: 6.6 mi stated against a 9.2 mi milepost range.
- UF campus extent conflicts: 68 mi or 20 mi.
- The Brightline hi-rail work almost certainly predates November 2023.

Recommendations

1. If ASPRS compliance matters, write it into the scope: at least 30 independent checkpoints, RMSE reporting per ASPRS Edition 2 v2 with the standard compliance statement, checkpoint survey error included, and LAS 1.4. Keep FDOT's Type A or B limits as well.
2. Start records requests with SR 570 Polk Parkway, the Keys municipal lidar and the City of Sarasota mapping. These are most likely to have sealed survey reports with checkpoint tables that can be recomputed as RMSE.
3. Treat any vendor "survey-grade" or "Type A" claim as unverified until you see the sealed survey report.